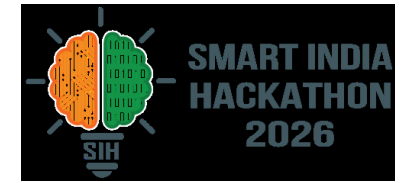
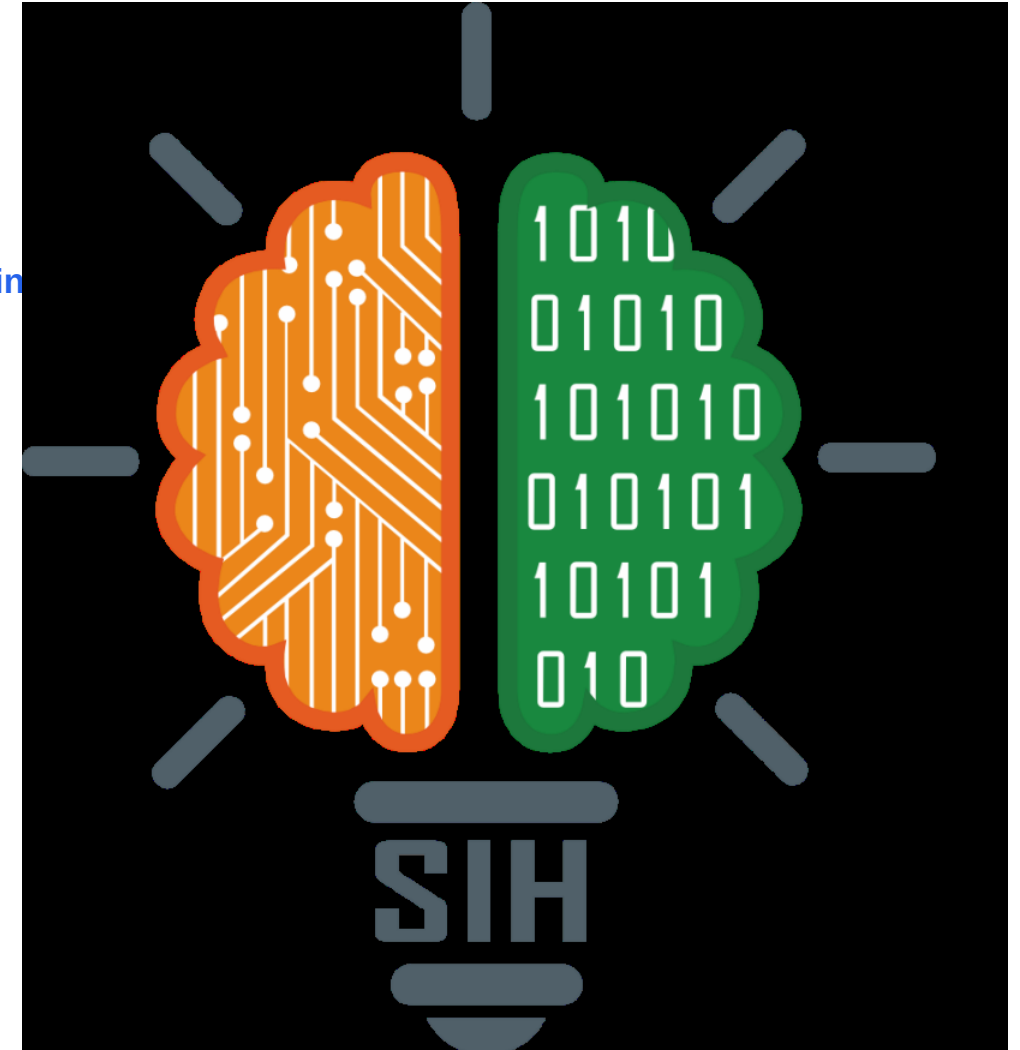


SMART INDIA HACKATHON 2026

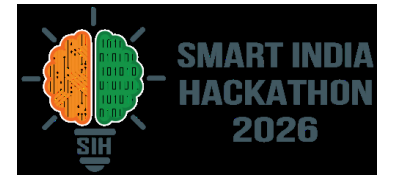


TITLE PAGE

- Problem Statement ID - [SIH26103](#)
- Problem Statement Title - [AI-Powered Predictive Risk Monitoring & Early-Warning](#)
for Centrally-Sponsored Infrastructure Projects
- Theme - Smart Automation / Infrastructure & Governance
- PS Category - Software
- Team ID - [\[Your Team ID\]](#)
- Team Name (Registered on portal) - [Your Team Name]



IDEA TITLE: INFRARISK MONITOR



◆ Proposed Solution (Describe your Idea/Solution/Prototype)

● Detailed explanation of the proposed solution:

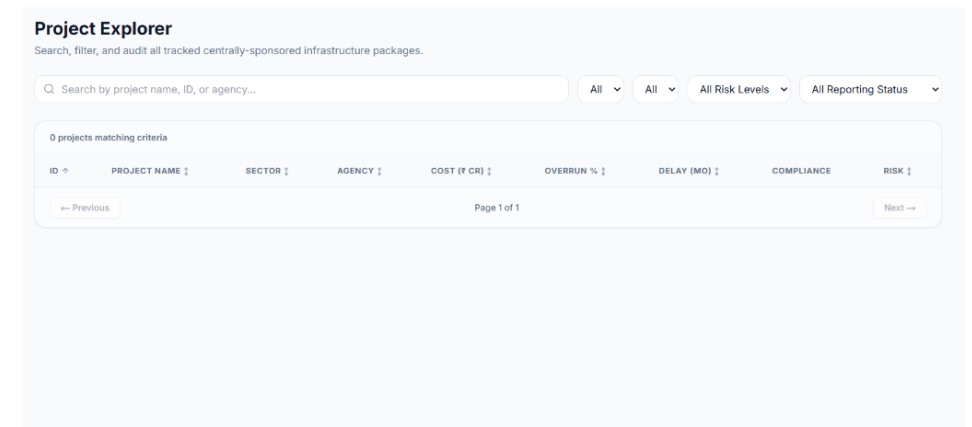
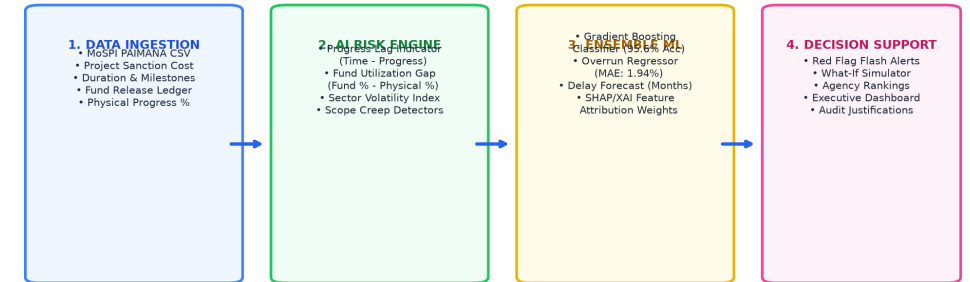
- Centrally deployed AI decision support platform built for MoSPI IPMD and Implementing Agencies.
- Continuously ingests PAIMANA project logs (expenditure, schedule, and physical milestones).
- Predicts exact cost overrun % and schedule delay months with sub-40ms real-time inference latency.

● How it addresses the problem:

- Replaces passive post-mortem tracking with anticipatory AI alerts months before budgets escalate.
- Identifies non-linear 'Progress Lag' (Elapsed% - Progress%) & 'Fund Gaps' that audits overlook.
- Includes an interactive What-If Risk Simulator allowing directors to test capital decisions live.

● Innovation and uniqueness of the solution:

- Dual ML Pipeline: Gradient Boosting Classifier (95.6% risk acc) + Regressor (1.94% MAE).
- Explainable AI (XAI): Transparent feature attribution weights for complete audit compliance.
- Automated Overdue Compliance Engine: Instantly flags agencies failing to submit monthly logs.

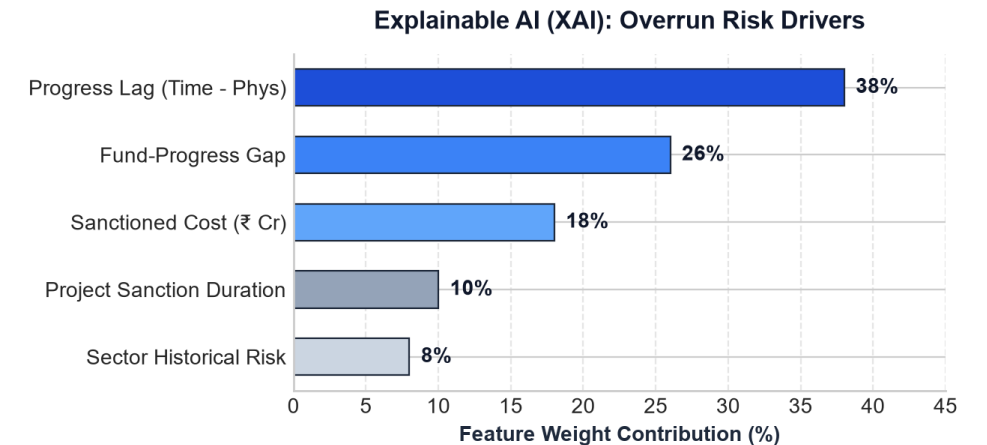
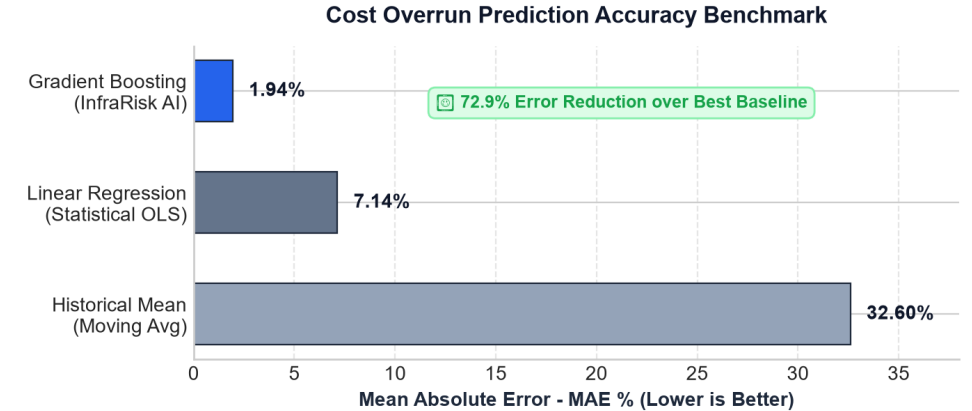


Technologies to be used:

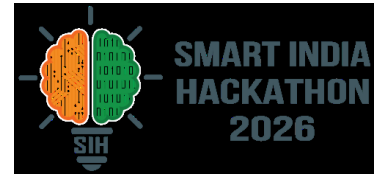
- Frontend Client: Next.js 14 (App Router), React 18, Tailwind CSS, Recharts (Responsive UI).
- Backend Microservices: Python 3.12, FastAPI, Uvicorn ASGI, Pydantic v2 schemas, RESTful APIs.
- ML & Analytics Core: Scikit-Learn Ensemble (Gradient Boosting), Joblib Serialization, Pandas, NumPy.
- Data Governance & Security: MoSPI PAIMANA schema, CSV staging engine, RBAC access control.

Methodology and Process for Implementation:

1. Ingestion & Preprocessing: Automated cleaning of project expenditure, timeline & milestone logs.
2. Feature Engineering: Derived indicators (Progress Lag = Elapsed% - Progress%; Fund Gap = Fund% - Progress%).
3. Dual Ensemble Modeling: Gradient Boosting Regressor (180 trees, max depth 5) + Classifier (140 trees).
4. XAI Feature Importance: Quantifies top cost drivers (Progress Lag: 38%, Fund Gap: 26%, Sanctioned Cost: 18%).
5. Dashboard & Alert Delivery: Real-time multi-channel feed, interactive what-if simulations, and agency ranking.



FEASIBILITY AND VIABILITY



- **Analysis of the Feasibility of the Idea:**

- Technical Feasibility: Working prototype operational; tested across 800 MoSPI mega-projects (<40ms latency).
- Operational Viability: Seamless integration with MoSPI PAIMANA schema; zero field workflow disruption.
- Financial Viability: 100% open-source core stack; zero recurring vendor licensing fees, minimal cloud compute.

- **Potential Challenges and Risks:**

- Data Incompleteness / Lag: Agency delays in updating monthly progress reports (>2 months overdue).
- Sector Heterogeneity: Disparate risk dynamics (Tunneling/Railways vs Solar Parks vs Expressways).
- Institutional Trust: Bureaucratic reluctance toward complex, opaque algorithmic scoring.

- **Strategies for Overcoming These Challenges:**

- Compliance Watchdog: Automated flagging of overdue reporting agencies directly on the executive overview.
- Sector-Encoded Stratification: Independent normalization weights for each sector to prevent cross-contamination.
- Explainable AI (XAI): Transparent SHAP feature attribution weights eliminating black-box mistrust.

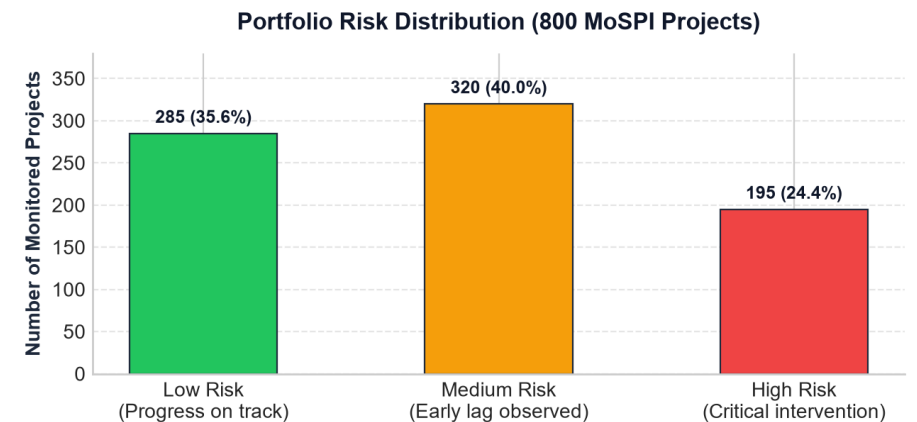
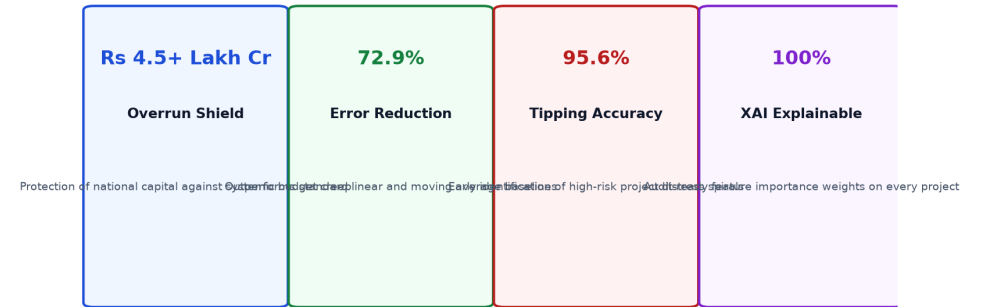
Technical Feasibility	Operational Viability	Economic Viability
Status: HIGH (Verified)	Status: SEAMLESS	Status: ZERO LICENSING
• End-to-end prototype operational • Sub-40ms inference latency • 95.6% risk category accuracy • Scalable microservice design	• Plugs into MoSPI PAIMANA schema • Zero change to field data entry • Browser & mobile accessible UI • Role-based Admin/Viewer access	• 100% open-source technologies • Low compute cloud footprint • High ROI: Saves Cr in overruns • Scalable to all 1800+ projects

● Potential Impact on the Target Audience:

- MoSPI & IPMD Central Monitoring: Transforms national project oversight from retrospective bookkeeping into proactive capital preemption across 1,800+ centrally-sponsored mega-projects.
- Implementing Agencies (NHAI, RVNL, NTPC, CPWD): Early identification of contractor distress spirals enabling corrective intervention before legal disputes and cost escalations become irreversible.
- Ministry Leadership & CCI: Provides empirical, data-backed justification for budgetary reallocation.

● Benefits of the Solution (Social, Economic, Environmental):

- Economic Benefit: Safeguards public capital against India's massive Rs 4.5+ Lakh Crore project cost overruns.
- Social Benefit: Accelerates timely commissioning of public utilities--hospitals, rural highways (PMGSY), metro rail transit corridors, and clean drinking water pipeline networks.
- Environmental Benefit: Minimizes construction waste, idle machinery emissions, and prolonged ecological disturbance.
- Governance Impact: Agency Accountability Index benchmark fosters competitive efficiency across PSUs.



- **Details / Links of the Reference and Research Work:**

- 1. Government Reports & Regulatory Frameworks**

- Ministry of Statistics and Programme Implementation (MoSPI), Govt. of India: Monthly Flash Reports on Central Sector Projects (Rs 150 Cr & Above), Infrastructure and Project Monitoring Division (IPMD).
 - MoSPI PAIMANA Portal: Project Assessment, Information Management & Analytical Network System (Official Project Attributes & Data Dictionary).
 - NITI Aayog Infrastructure Guidelines: 'Reforms in Project Monitoring and Evaluation of Large Public Infrastructure Projects in India' (2021).

- 2. Academic Research & Theoretical Foundations**

- Flyvbjerg, B., Holm, M. K. S., & Buhl, S. L. (Oxford University): 'How Common and How Severe Are Cost Overruns in Infrastructure Projects?' - Project Management Journal (Empirical analysis of non-linear escalation dynamics).
 - Friedman, J. H. (Stanford University): 'Greedy Function Approximation: A Gradient Boosting Machine' - The Annals of Statistics (Theoretical foundation for non-linear regression ensembles).
 - Lundberg, S. M., & Lee, S. I. (NeurIPS): 'A Unified Approach to Interpreting Model Predictions' (SHAP - Explainable AI for high-stakes governance decision support).

- 3. Verified Production Artifacts & Prototype Codebase**

- Full Open-Source Codebase & ML Models: https://github.com/himanshux18/SIH_Project.git
 - Calibrated 800-Project Infrastructure Dataset: PAIMANA-compliant synthetic dataset with verified empirical cost & progress lag distributions.
 - Benchmarked Performance: Gradient Boosting Regressor (1.94% MAE) & Classifier (95.6% Accuracy) beating standard OLS Linear Regression by +72.9%.